

XII TABLES FOR RESEARCHERS ON EXTRATERRESTRIAL INTELLIGENCE

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Abstract—In the legal field, studies on extraterrestrial intelligence began four years before the leading article on interstellar communication was published. Even though discussion of the legal problems still lacks a juridical framework of universal acceptance, a few guidelines are suggested under the title of XII Tables (or Commandments). The XII Tables, which was the Romans' first code, also included provisions on international law and may be seen as a very important source of the law of nations. Their great merit was brevity, which allowed the progressive development of the law. These XII Tables have been divided into six to guide reactions in the event messages or signals are received, and six to determine the behavior of those responsible for answering if and when a communication attempt takes place. They were drafted on the basis of what is known as "ethics" and "law" in a way that any progress in this respect may be to the credit of our civilization and the human condition. For a wise and cautious analysis of the matter, the creation of a committee of representatives of governmental and non-governmental organizations is proposed.

1. CONCERN OF JURISTS

Space law had forerunners of its current legal solutions before technical developments and even before scientific research took place. Vladimir Mandl, a Czech jurist, published *Das Weltraumrecht: Ein Problem der Raumfahrt* in 1932, where he answered questions which seemed totally extraneous to jurists of his generation. Gifted legal researchers acted as imaginative scientists and authors in anticipating technological activities in outer space and on celestial bodies.

1.1 *The Work of Andrew G. Haley (from 1955)*

Four years before the leading article by G. Cocconi and Philip Morrison "Searching for Interstellar Communications,"[1] Andrew G. Haley, a well-known lawyer who presided over the International Aeronautical Federation (IAF) and its Permanent Legal Committee, introduced to public audiences in 1955 *Basic concepts of Space Law* at the 25th Anniversary Meeting of ARS in Chicago, and on June 7, 1955 *Jurisdiction Beyond the Earth* at the Charlotte, N. C. Rotary Club. Subsequently, he presented on September 19, 1956 at the VII IAF congress *Space Law and Metalaw - A Synoptic View*, and in a special supplement of *The Harvard Law Review*[2] a study on "Recent Development in Space Law and Metalaw," etc.

I was closely connected to Haley and enjoyed his friendship and his hospitality at his home in Washington up to his death in 1966. Upon my invitation he visited Buenos Aires twice, and we worked together within the Permanent Legal Committee of the IAF and the International Institute of Space Law (IISL). On many an

occasion he talked about metalaw. But we wrote no joint paper on the subject, because I was convinced that studies, which he published regularly, and Chapter 12 of his book *Space Law and Government*[3] were sufficient at the time. Ernst Fasan analyzed his work in depth and we will therefore make no further reference to it at this point.

1.2 *The Book of Ernst Fasan (1970)*

The subject was taken up with merit and dedication by Ernst Fasan, a distinguished Austrian jurist, who authored the one existing book to date on *Relations with Alien Intelligences*[4]. I will not comment on this book nor on his latter papers here, but I only wish to mention that he is very active in both the Academy and the Institute, and very productive as well.

1.3 *Enquiry by Dr. G. C. M. Reijnen (1976)*

A survey was addressed to sixteen jurists, diplomats and scientists by Gelberta Cornelia Maria Reijnen, from Netherlands, who then published the results in her doctoral thesis on *Legal Aspects of Outer Space*[5]. This survey was greatly stimulating for all concerned, including myself.

1.4 *The Innsbruck Symposium (1986)*

On October 9, 1986, the CETI Committee of the International Academy of Astronautics (IAA) conducted a session on "The Legal, Political and Social Implications of the

Detection of an Extraterrestrial Intelligent Signal." The invitation to participate in said symposium, read: "This is a topic that will require a large amount of liaison between the scientific and legal communities in preparation for any large scale systematic search efforts and after the (hopeful) detection of an extraterrestrial signal." On that occasion, I stated orally a number of principles which I call the XII Tables and I undertook to draft a text for the Brighton meeting in 1987.

2. ARGENTINE CONTRIBUTION

2.1 *Teaching of the Subject (Buenos Aires, since 1960 and 1962)*

Both the chair of Air and Space Law of the Salvador University of Buenos Aires created in 1960 (the first in the world where the study of Space Law is mandatory in order to obtain all law degrees) and the National Institute of Air and Space Law (a Graduate Academic Institution with the rank of National University since 1962) which has the one independent chair of Space Law, devoted a lecture in one of their courses to metalaw or legal relations with extraterrestrial intelligence.

2.2 *The Round Table of Guayaquil (1963)*

After inaugurating the Space Law courses at the University of Quito and the University of Guayaquil, the head of the latter asked me to organize and to conduct a round table on the subject in Guayaquil, Ecuador. This meeting took place on May 26, 1963 and, cautiously, I suggested the following title: "Has Mankind Any Law-Making Power Beyond Its Planet?" I will not dwell on individual statements, but only on the conclusions. The answer was very clear: Yes, mankind has law-making power beyond its planet. As far as the possibility of contacts with other intelligences is concerned, in the large framework of natural law, the Round Table reserved the right to give its opinion until the eventual intelligent beings' attitude to what we call law becomes known.

2.3 *The Lecture of Eugen Pepin (Buenos Aires, 1964)*

In 1964 in Buenos Aires the International Institute of Space Law held the first round table

which was intended to consider legal rules pertaining to satellite communications. Its President and Chairman of the meeting, Dr. E. Pepin, also gave a lecture on "The Future of Space Law," in which he suggested the drafting a conduct code for eventual ultra-human contacts, on the part of scholars and astronauts.

2.4 *The Foreword to Ernst Fasan's Book (1970)*

I had the privilege of writing one of the forewords to the book by Ernst Fasan. Therein I said: "The problem of relations with extraterrestrial intelligences is also a legal question. Such is the point of view of those who consider law a science of solutions which, in order to be fully efficient, have to precede facts. We consider it to be the bridge between principles and practice. The space jurists have proved this by establishing rules, those of the 1967 Space Treaty, for an area unknown to them: Man extended his law-making power to an immense zone into which he has just peering and to celestial bodies about which he knows little. However, outer space law covers both areas entirely[6]."

2.5 *Symposium of Montevideo (1982)*

In commemoration of UNISPACE 82, the Uruguayan government organized an international symposium on 3-4 June 1982 in Montevideo on the "Present Problems of the Cosmos." One session was devoted to extraterrestrial life. Scientific progress in the exploration of the cosmos was examined as were the legal aspects. I was entrusted with the analysis and updating of knowledge on extraterrestrial legal problems.

2.6 *International Meeting on "Intelligent Life in the Universe" (Buenos Aires, 1985)*

Students of Physics at the School of Exact and Natural Science at the University of Buenos Aires organized between December 20 and December 22, 1985, an interdisciplinary meeting on "Intelligent Life in the Universe," which was attended by outstanding members of the international scientific community in the field. It was a success and the public filled up the main hall of the School. I spoke on the legal issues implied in extraterrestrial relations, as the last speaker, late on December 21, 1985. Exchanges

with the students and the public were lively, since they were very interested in knowing more about the person of Philip Morrison. I had worked with Dr. Morrison in New York between March 9-10, 1982 on the subject of the human condition and the alternatives it faces on account of future space developments. Time was up, the meeting was closed and the questions kept coming up, even after the light was turned off. Another result of the meeting was the creation of the magazine *Astrofisica*, now regularly published under the aegis of the Instituto de Astronomia y Fisica del Espacio of the University of Buenos Aires and the Consejo Nacional de Investigaciones Cientificas y Tecnicas (CONICET).

3. XII TABLES FOR RESEARCHERS

XII Tables are an attempt to present general principles rich in legal content in a brief code. I have called them by this title as a tribute to the earliest Roman code, internationally compiled at first and universally current afterwards.

I used the expression for the first time to refer to guidelines when I was analyzing legal issues related to Direct Broadcasting by Satellites (DBS) at the United Nations, in 1970. These principles were finally incorporated into UNGA Resolution 37/42, December 10, 1982. Encouraged by the acceptance of this legal denomination I drafted another XII Tables for Solar Energy in 1975, which was adopted at national and regional fora and eventually by the Interamerican Bar Association in its Conference of Quito (1981). In drafting the XII Tables for Researchers on Extraterrestrial Intelligence I have tried to sum up as many proposals as possible with the necessary focus on their legal and ethical character. During the question period following the aforementioned 1964 lecture by Eugene Pepin, it was said that jurists should provide the legal and ethical framework for researchers and astronauts, bearing in mind that the latter are the envoys of mankind.

THE XII TABLES

A. Reactions in case of receiving messages or signals:

I. Principle of international report and consultation. If a researcher receives messages or signals, he must:

a) Report the fact in consultation with an International Extraterrestrial Communication Committee. Such Committee should be made up by representatives of:

- the United Nations Committee on the Peaceful Uses of Outer Space (COPUOS)
- the International Telecommunications Union
- the International Astronomical Union
- the International Council of Scientific Unions Committee on Space Research (COSPAR)
- the International Academy of Astronautics
- the International Institute of Space Law

b) The report must be made immediately and must not necessarily prompt a meeting of the Committee.

II. Principle of caution. All persons concerned must abstain from answering until the Committee opinion is known.

III. Principle of prior checking. Requests from other observatories and scientific centers active in extraterrestrial intelligence research will be made for assistance to confirm the veracity of the attempt to communicate by extraterrestrials.

IV. Principle of truthfulness. The addressee of messages or signals must never be intentionally deceived.

V. Principle of truthfulness. Researchers must always observe due loyalty vis-a-vis all other researchers.

VI. Principle of prompt reply. Results of consultations must be communicated as soon as technically possible.

B. Conduct to be observed by researchers and those responsible for answering:

VII. Principle of universal answer. The answer will be made in the name and on behalf of mankind, not of nations, or governments or the international community.

VIII. Principle of friendly welcome. A peaceful and friendly welcome must be given and every effort to understand extraterrestrials must be made.

IX. Principle of understanding and non-aggression. Every effort towards understanding extraterrestrials and towards a non-hostile attitude must be made.

X. Principle of cooperation. Cooperation in the scientific and the technological fields, including the arts and social sciences, shall be offered.

XI. Principle of respect. The habits, customs, manners, practices, languages, religions or creeds, organizational systems and styles of life must be respected at all times.

XII. Principle of preservation of life and property. All forms of life must be preserved from any kind of damage or intrusion, i.e. the illegal entering or taking of an extraterrestrial's property.

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4. E. Fasan, *Relations with Alien Intelligences*. Berlin (1970).
5. G. C. M. Reijnen, *Legal Aspects of Outer Space*. Ph.D. Dissertation (1976).
6. Foreword to reference 4.